

B-9-Y

Roll No.

Total No. of Questions : 4]

[Total No. of Printed Pages : 8

12thARF(SZ)JKUT2024-25

309-Y

CHEMISTRY

Time : 3 Hours]

[Maximum Marks : 70

General Instructions :

- (i) There are total four Sections in the question paper. All questions are compulsory.
- (ii) **Section-A** contains 10 Objective Type Questions (Multiple Choice Questions) of 1 mark each. $1 \times 10 = 10$ marks
- (iii) **Section-B** contains 9 Very Short Answer Type Questions of 2 marks each to be answered in 20-30 words. $2 \times 9 = 18$ marks
- (iv) **Section-C** contains 9 Short Answer Type Questions of 3 marks each to be answered in 100-150 words. $3 \times 9 = 27$ marks
- (v) **Section-D** contains 3 Long Answer Type Questions of 5 marks each to be answered in 150-200 words. $5 \times 3 = 15$ marks
- (vi) Use log table if necessary. Use of scientific calculators is not allowed.

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SECTION-A

1 each

OBJECTIVE TYPE QUESTIONS
(MULTIPLE CHOICE QUESTIONS)

1. Select the correct one :

(i) Which one of the following is not the correct unit of molal elevation constant ?

(A) $^{\circ}\text{C}/\text{m}$

(B) K/m

(C) K kg mol^{-1}

(D) K mol kg^{-1}

(ii) Which of the following solutions will have the maximum lowering of vapour pressure at 300 K ?

(A) 1 m NaCl

(B) 1 m BaCl_2

(C) 1 m Sucrose

(D) 1 m Phenol

(iii) The electrode potential of a half cell depends upon :

- (A) Nature of metal
- (B) Concentration metal ions in solution
- (C) Temperature
- (D) All the three above

(iv) The rate constant of a reaction has same units as the rate of reaction. The reaction is of :

- (A) Zero order
- (B) First order
- (C) Second order
- (D) None of these

(v) S_N^2 mechanism proceeds through intervention of :

- (A) Carbonium ion
- (B) Transition state
- (C) Free radical
- (D) Carbanion

(vi) Phenol on reduction with H_2 in presence of nickel catalyst gives :

- (A) Benzene
- (B) Toluene
- (C) Cyclohexane
- (D) Cyclohexanol

(vii) Which of the following gives carbylamine reaction ?

- (A) CH_3NH_2
- (B) $C_2H_5NO_2$
- (C) CH_3CONH_2
- (D) $(CH_3)_2NH$

(viii) Ligands are :

- (A) Negative ions only
- (B) Neutral molecules
- (C) Never positive ions

(D) Negative or positive ions or neutral molecules

(ix) Which of the following constitutes the genetic material of the cell ?

(A) Nucleic acid

(B) Proteins

(C) Lipids

(D) Carbohydrates

(x) Vitamin A is called :

(A) Ascorbic acid

(B) Retinol

(C) Calciferol

(D) Tocopherol

SECTION-B

2 eac

VERY SHORT ANSWER TYPE QUESTIONS

2. (i) A first order reaction is found to have a rate constant $k = 5.5 \times 10^{-4} \text{ s}^{-1}$. Find the half-life period.

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- (ii) How many transition series of elements are there in the periodic table ? Name them.
- (iii) What is ammonolysis ? Give example.
- (iv) What is rectified spirit ?
- (v) What are essential and non-essential amino acids ?
- (vi) Write the IUPAC names of :
 - (a) $\text{Zn}_2[\text{Fe}(\text{CN})_6]$
 - (b) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]^{2+}$
- (vii) Define rate of reaction and rate constant.
- (viii) Why cannot molecularity of any reaction be equal to zero ?
- (ix) Give a brief account of Wurtz reaction.

SECTION-C

3 each

SHORT ANSWER TYPE QUESTIONS

3. (i) Aldehydes and ketones undergo nucleophilic addition reactions. Justify with examples.
- (ii) Define Kohlrausch's law. How can it be used to find the degree of dissociation of a weak electrolyte ?

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- (iii) Explain, why transition metal ions usually show paramagnetic behaviour ?
- (iv) What happens when acidified KMnO_4 reacts with (a) H_2S
(b) Oxalic acid (c) KNO_2 ?
- (v) Define cis and trans isomerism. Draw the cis and trans isomers of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ ion. <https://www.jkboseonline.com>
- (vi) How will you convert ethyl bromide to ?
- (a) Ethoxyethane
- (b) Ethane
- (c) Ethanol
- (vii) How will you distinguish between primary, secondary and tertiary alcohols by Victor Meyer's test ? Give chemical reactions.
- (viii) What is Diazotisation ? Discuss its mechanism.
- (ix) What are Carbohydrates ? How are they classified ?

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SECTION-D

5 each

LONG ANSWER TYPE QUESTIONS

4. (i) State and explain Raoult's law for (a) Volatile solute (b) non-volatile solute.

Or

What do you understand by Colligative properties of a solution ?
Explain briefly osmosis and osmotic pressure.

- (ii) What is an electrochemical cell ? Describe working of Daniell cell.

Or

What is Corrosion ? Give mechanism (electrochemical phenomenon) of rusting of iron. What do you understand by cathodic and barrier protection of corrosion ?

- (iii) Describe any *five* general methods of preparations of carboxylic acids.

Or

Giving chemical equations write a brief account of the following :

- (a) Aldol condensation
- (b) Cannizzaro reaction
- (c) Rosenmund reduction